

Azhari Abbas

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EDUCATION

- **Embry-Riddle Aeronautical University** Daytona Beach, Florida
Bachelor of Science — Aerospace Engineering (GPA: 3.88) August 2022 — May 2026
- **Relevant Coursework:** Attitude Dynamics & Controls, Space & Rocket Propulsion, Aerodynamics, Material Science, Structural Mechanics, Spacecraft Systems, Spacecraft Design, Orbital Mechanics, Computer Science, Electrical Engineering Fundamentals

SKILLS

- **Computer Aided Design:** CATIA, SolidWorks, AutoCAD, Siemens NX, Autodesk Inventor, Fusion 360
- **Simulation and Analysis:** Simulink, LabVIEW, NASTRAN, Ansys, Thermal Desktop, STK, GMAT, MES, CPLM
- **Programming and Flight Software:** Python, MATLAB, C, C++, JavaScript, CUDA, SQL, Fortran, Arduino, ROS, PX4

EXPERIENCE

- **Mitsubishi Power Americas Inc.** Orlando, FL
Associate, Performance Engineering & Product Management (Part-time) April 2024 - May 2026
 - Analyzed operational data from high-capacity gas turbines to devise product life cycle management processes, support the development of new product strategies, and assist in crafting business plans.
 - Managed day-to-day communication between engineering, sales, and program managers throughout prioritization, strategy, production, and process improvement cycles.
- **Northrop Grumman Corporation** Palmdale, CA
Test and Evaluation Engineering Intern (Full-time) June 2025 — August 2025
 - Active U.S. Department of Defense Secret/SAP Clearance.
 - Responsible for the comprehensive testing and evaluation of hydraulic, environmental control, electrical, and fuel systems, together with writing Acceptance Test Procedures (ATPs) to ensure system compliance and safety.
 - Refined augmented reality (AR) software frameworks and tools with real-time instructions to guide technicians and engineers during system testing procedures.
- **Docking Robotics LLC** Daytona Beach, FL
Mechanical Engineering Intern (Part-time) June 2023 — November 2023
 - Performed mechanical installation design and integration of encoders for an autonomous ground vehicle, including the development of custom hardware and mounting solutions.
 - Collaborated with vendors to troubleshoot and resolve measurement system challenges, optimizing component selection and ensuring reliable instrumentation performance.

PROJECTS

- **OrbitFlex CubeSat Team** August 2025 — Present
Environmental and Test Engineer
 - Developed end-to-end testing plans for a 3U CubeSat aligned with NASA GEVS vibration, shock, and functional integrity requirements to validate structural and operational readiness for GEO SSA missions.
 - Supported contamination control and cleanroom operations, implementing ISO Class 8 protocols and MIL-STD-1246C surface cleanliness standards to ensure mission reliability and compliance.
- **Navigating Enhanced Exploration Assistance (NEXA)** January 2023 — Present
Principal Investigator (NASA Engineering Physics Propulsion Lab)
 - Developed an AI-powered autonomous drone security system for real-time surveillance, integrating computer vision with drone hardware for threat detection, and autonomous navigation.
 - Implemented Ultralytics YOLO based object detection deployed on NVIDIA Jetson embedded systems for low-latency edge inference, enabling real-time alerts via high-resolution imaging and secure communication modules.
- **ESA SPARK AI Challenge - Spacecraft Detection & Segmentation** January 2026 — March 2026
Undergraduate Researcher
 - Developed a multi-task deep learning model for spacecraft classification, detection, and segmentation in synthetic orbital imagery with efficient architecture balancing accuracy and computational cost (FLOPs & parameters).
 - Achieved Top 5% ranking with 93.98% classification accuracy and 0.927 detection IoU, while implementing multi-head segmentation for spacecraft body and solar panels under challenging conditions (occlusion, glare, and scale variation).
- **Attitude Dynamics and Control of a Rigid Spacecraft in LEO** May 2025 — June 2025
Project Lead
 - Led the end-to-end simulation of spacecraft attitude dynamics in MATLAB using quaternion-based formulations, modeling free rigid-body motion, gravity-gradient effects, and active stabilization scenarios in Lower Earth Orbit (LEO).
 - Designed and tuned a diagonal PD controller to achieve critical damping and global asymptotic stability, validating performance through numerical integration of quaternion-rate dynamics and control torque analysis.
- **EUREKA Rocket Launch Team** January 2023 — October 2023
Programming Lead
 - Led development of avionics and data logging systems for a 5-ft LASERHAWK rocket, enabling GPS-based flight tracking and trajectory reconstruction, contributing to 1st place at the NSBE-Boeing Rocket Launch Competition.
 - Designed and implemented Arduino/C++ embedded software for real-time sensor acquisition and data synchronization, optimizing reliability under high-vibration launch conditions and supporting post-flight performance analysis.